

Log Correlation Agent (Agent 8)

Deep Technical Specification & Operational Manual

Executive Overview & Purpose

The Log Correlation Agent performs deterministic time-series correlation across firewall, IDS/IPS, authentication, and endpoint event streams to detect multi-stage attack chains that individual log entries would miss.

1. Architectural Overview & Workflow

- Position in LangGraph Engine: Third node in Pipeline B, executing after IOCEnrichmentAgent.
- Input Consumption: Consumes raw log payloads, enriched IOCs, and triage context.
- State Propagation: Writes `state.log_correlation_report` with correlated event clusters and attack-chain hypotheses.

2. Deterministic Correlation Engine

- Source IP Clustering: Groups events sharing a source address within a configurable time window.
- Port/Service Chaining: Maps sequential port-scan to service-access to privilege-escalation patterns.
- Authentication Failure Counting: Flags brute-force attempts exceeding deterministic thresholds (e.g., >5 failures/minute).
- Temporal Window Analysis: Only events inside the correlation window are chained, preventing false-positive merging.

3. Attack-Chain Hypothesis Generation

- Chain Reconstruction: Each correlated cluster produces a human-readable hypothesis of the likely attack flow.
- Confidence Scoring: Chain confidence reflects evidence volume and consistency across independent log sources.
- Alert Fatigue Reduction: Correlated clusters are surfaced once with full evidence rather than as disjointed alerts.